

Lab 5 Activity

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.5
v forcats    1.0.1      v stringr    1.5.2
v ggplot2    4.0.0      v tibble     3.3.0
v lubridate  1.9.4      v tidyr      1.3.1
v purrr      1.1.0
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

1

Suppose that you are planning to conduct an experiment to compare a new treatment to a placebo, and you want to be able to detect a relatively small effect ($d = 0.3$) with $\alpha = .05$. You have the budget to recruit $N = 100$ subjects for your study. Based on power considerations alone, would you prefer to conduct a paired-samples t -test or an independent-samples t -test (with equally sized groups)? Do the power calculations in R. For the test with less power, how many additional subjects would you need to recruit to have the same power as the more powerful test?

```
library(pwr)
# for a two-tailed test:

pwr.t.test(n = 100, d = .3, sig.level = .05, power = NULL,
           type = "paired", alternative = "two.sided") # power = .84
```

Paired t test power calculation

```
n = 100
d = 0.3
sig.level = 0.05
power = 0.8439471
alternative = two.sided
```

NOTE: n is number of *pairs*

```
pwr.t.test(n = 50, d = .3, sig.level = .05, power = NULL, # remember: is the number in each
           type = "two.sample", alternative = "two.sided") # power = .32
```

Two-sample t test power calculation

```
n = 50
d = 0.3
sig.level = 0.05
power = 0.3178022
alternative = two.sided
```

NOTE: n is number in *each* group

```
pwr.t.test(n = NULL, d = .3, sig.level = .05, power = .8439471,
           type = "two.sample", alternative = "two.sided")
```

Two-sample t test power calculation

```
n = 197.0875
d = 0.3
sig.level = 0.05
power = 0.8439471
alternative = two.sided
```

NOTE: n is number in *each* group

```
# n = 197 in each group - nearly 4x as many as the original!

# for a one-tailed test:

pwr.t.test(n = 100, d = .3, sig.level = .05, power = NULL,
           type = "paired", alternative = "greater") # power = .91
```

Paired t test power calculation

```
      n = 100
      d = 0.3
sig.level = 0.05
power = 0.9089875
alternative = greater
```

NOTE: n is number of *pairs*

```
pwr.t.test(n = 50, d = .3, sig.level = .05, power = NULL, # remember: is the number in each
           type = "two.sample", alternative = "greater") # power = .44
```

Two-sample t test power calculation

```
      n = 50
      d = 0.3
sig.level = 0.05
power = 0.4383254
alternative = greater
```

NOTE: n is number in *each* group

```
pwr.t.test(n = NULL, d = .3, sig.level = .05, power = .9089875,
           type = "two.sample", alternative = "greater")
```

Two-sample t test power calculation

```
      n = 197.9428
      d = 0.3
```

```

sig.level = 0.05
power = 0.9089875
alternative = greater

```

NOTE: n is number in *each* group

```
# n = 197 in each group - nearly 4x as many as the original!
```

2

For the independent samples *t*-test, to what extent does having equally vs. unequally sized groups affect power? Test for a two-tailed test using $d = .5$, $N = 100$, and $\alpha = .05$. At what group sizes will power be less than .5?

```
pwr.t2n.test(n1 = 50, n2 = 50, d = .5, sig.level = .05, power = NULL, alternative = "two.sided")
```

t test power calculation

```

n1 = 50
n2 = 50
d = 0.5
sig.level = 0.05
power = 0.6968934
alternative = two.sided

```

```
pwr.t2n.test(n1 = 25, n2 = 75, d = .5, sig.level = .05, power = NULL, alternative = "two.sided")
```

t test power calculation

```

n1 = 25
n2 = 75
d = 0.5
sig.level = 0.05
power = 0.5729551
alternative = two.sided

```

```
pwr.t2n.test(n1 = 15, n2 = 85, d = .5, sig.level = .05, power = NULL, alternative = "two.sided")
```

```
t test power calculation
```

```
      n1 = 15
      n2 = 85
      d = 0.5
sig.level = 0.05
power = 0.4239257
alternative = two.sided
```

```
pwr.t2n.test(n1 = 20, n2 = 80, d = .5, sig.level = .05, power = NULL, alternative = "two.sided")
```

```
t test power calculation
```

```
      n1 = 20
      n2 = 80
      d = 0.5
sig.level = 0.05
power = 0.5081857
alternative = two.sided
```

```
pwr.t2n.test(n1 = 19, n2 = 81, d = .5, sig.level = .05, power = NULL, alternative = "two.sided")
```

```
t test power calculation
```

```
      n1 = 19
      n2 = 81
      d = 0.5
sig.level = 0.05
power = 0.4929878
alternative = two.sided
```

3

How does changing the degrees of freedom in a chi-square test affect power, holding other factors constant? Answer this question by trying out various calls to `pwr.chisq.test` in R. How might this affect your experimental design?

```
pwr.chisq.test(w = .5, N = 50, df = 1, sig.level = .05)
```

Chi squared power calculation

```
w = 0.5  
N = 50  
df = 1  
sig.level = 0.05  
power = 0.9424375
```

NOTE: N is the number of observations

```
pwr.chisq.test(w = .5, N = 50, df = 2, sig.level = .05)
```

Chi squared power calculation

```
w = 0.5  
N = 50  
df = 2  
sig.level = 0.05  
power = 0.8962428
```

NOTE: N is the number of observations

```
pwr.chisq.test(w = .5, N = 50, df = 3, sig.level = .05)
```

Chi squared power calculation

```
w = 0.5  
N = 50  
df = 3  
sig.level = 0.05  
power = 0.8561735
```

NOTE: N is the number of observations

```
pwr.chisq.test(w = .1, N = 500, df = 1, sig.level = .05)
```

Chi squared power calculation

```
w = 0.1  
N = 500  
df = 1  
sig.level = 0.05  
power = 0.6087795
```

NOTE: N is the number of observations

```
pwr.chisq.test(w = .1, N = 500, df = 2, sig.level = .05)
```

Chi squared power calculation

```
w = 0.1  
N = 500  
df = 2  
sig.level = 0.05  
power = 0.5036664
```

NOTE: N is the number of observations

```
pwr.chisq.test(w = .1, N = 500, df = 3, sig.level = .05)
```

Chi squared power calculation

```
w = 0.1  
N = 500  
df = 3  
sig.level = 0.05  
power = 0.440509
```

NOTE: N is the number of observations

adding degrees of freedom (categories) lowers power - may want to include fewer categories